

Critical appraisal (sort of)

Tengku Muhammad Hanis Bin Tengku Mokhtar

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Outline

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- [AI tools - use or not?](#)

Why read a research paper?

- It's your job
- Update the knowledge on the progress in the field of study
- To get an idea to carry out research works
- Find support on one's view
- Understand certain fundamental aspect (eg; pathophysiology of disease)
- To know certain aspect of one's research (eg; causation, clinical feature, etc)

Types of research paper

- Original research article:
 - Very common
 - Standard format: IMRaD (intro, methods, results, and discussion)
- Review article:
 - Provide a comprehensive summary of research on a certain topic
 - Types:
 - Narrative review/non-systematic review
 - Systematic review
 - Scoping review
 - Bibliometric review
 - Meta-analysis
- Conference proceeding article/abstract:
 - Presented at conference and usually less comprehensive/detail

- Preprint article:
 - Non-peer reviewed article that is shared publicly
- Editorial
 - Opinion or view of editorial board member (usually) - an expert
 - Reflect the stance of the journal
- Case report/case series:
 - Report specific and interesting phenomenon/occurrence
 - More common in medicine
- Short reports/letters:
 - Communicate brief reports of data from original research that editors believe will be interesting to many researchers
 - May indicate future research
- Etc

BMJ Open Net survival differences of breast cancer between stages at diagnosis and age groups in the east coast region of West Malaysia: a retrospective cohort study

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ABSTRACT

Objective Estimation of the net survival of breast cancer helps in assessing breast cancer burden at a population level. Thus, this study aims to estimate the net survival of breast cancer at different cancer staging and age at diagnosis in the east coast region of West Malaysia.

Setting Kelantan, Malaysia.

Participants All breast cancer cases diagnosed in 2007 and 2011 identified from Kelantan Cancer Registry.

Design This retrospective cohort study used a relative survival approach to estimate the net survival of patients with breast cancer. Thus, two data were needed: breast cancer data from Kelantan Cancer Registry and general population mortality data for Kelantan population.

Primary and secondary outcome measures Net survival according to stage and age group at diagnosis at 1, 3 and 5 years following diagnosis.

Results The highest net survival was observed among stage I and II breast cancer cases, while the lowest net survival was observed among stage IV breast cancer cases. In terms of age at diagnosis, breast cancer cases aged 65 and older had the best net survival compared with the other age groups.

Conclusion The age at diagnosis had a minimal impact on the net survival compared with the stage at diagnosis. The finding of this study is applicable to other populations with similar breast cancer profile.

INTRODUCTION

Breast cancer is the most common cancer

Strengths and limitations of this study

- This population-based study used state cancer registry data to describe the net survival of breast cancer in the east coast region of West Malaysia.
- The use of relative survival approach to estimate the net survival is considered as a standard practice for a population-based study.
- This study expanded an abridged life table of the general population mortality data into a complete life table to estimate the net survival due to unavailability of a complete life table.
- The use of different method to expand the mortality data leading to a lack of standardisation in this research area.
- This study lacked socioeconomic, sociodemographic and clinical information which may further explain our finding.

8, respectively.³ In terms of breast cancer survival, a 5-year relative survival between 2005 and 2009 in Malaysia was 67.8%, while other Asian countries such as South Korea (82.7%), Indonesia (77.7%) and Thailand (71.3%) had a better survival.⁴ Breast cancer staging and age are important prognostic factors of breast cancer. Early-stage patients with breast cancer are expected to have better



pharmaceuticals

Review

Periodontal Wound Healing and Tissue Regeneration: A Narrative Review

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Abstract: Periodontal disease is a major public health issue, and various periodontal therapies have been performed to regenerate periodontal tissues. The periodontium is a complex structure composed of specialized tissues that support the teeth, and most periodontal surgeries are invasive procedures, including a resection of the gingiva or the alveolar bone. The periodontal wound healing process is slightly different from cutaneous wound healing and is similar to fetal healing, being almost scar-free. The aim of this review article is to provide an overview of periodontal wound healing and discuss various surgical and pharmaceutical approaches to achieve stable wound healing and improve the treatment outcomes. In addition, detrimental and limiting factors that induce a compromised prognosis are discussed, along with the perspective and future direction for successful periodontal tissue regeneration.

Keywords: periodontium; periodontal tissue regeneration; wound healing



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RESEARCH ARTICLE

The burden of premature mortality from cardiovascular diseases: A systematic review of years of life lost

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Abstract

Introduction

Premature mortality refers to deaths that occur before the expected age of death in a given population. Years of life lost (YLL) is a standard parameter that is frequently used to quantify some component of an "avoidable" mortality burden.

Objective

To identify the studies on premature cardiovascular disease (CVD) mortality and synthesise their findings on YLL based on the regional area, main CVD types, sex, and study time.

Method

We conducted a systematic review of published CVD mortality studies that reported YLL as an indicator for premature mortality measurement. A literature search for eligible studies was conducted in five electronic databases: PubMed, Scopus, Web of Science (WoS), and the Cochrane Central Register of Controlled Trials (CENTRAL). The Newcastle-Ottawa

Diagnostic Accuracy of Machine Learning Models on Mammography in Breast Cancer Classification: A Meta-Analysis

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Abstract: In this meta-analysis, we aimed to estimate the diagnostic accuracy of machine learning models on digital mammograms and tomosynthesis in breast cancer classification and to assess the factors affecting its diagnostic accuracy. We searched for related studies in Web of Science, Scopus, PubMed, Google Scholar and Embase. The studies were screened in two stages to exclude the unrelated studies and duplicates. Finally, 36 studies containing 68 machine learning models were included in this meta-analysis. The area under the curve (AUC), hierarchical summary receiver operating characteristics (HSROC) curve, pooled sensitivity and pooled specificity were estimated using a bivariate Reitsma model. Overall AUC, pooled sensitivity and pooled specificity were 0.90 (95% CI: 0.85–0.90), 0.83 (95% CI: 0.78–0.87) and 0.84 (95% CI: 0.81–0.87), respectively. Additionally, the three significant covariates identified in this study were country ($p = 0.003$), source ($p = 0.002$) and classifier ($p = 0.016$). The type of data covariate was not statistically significant ($p = 0.121$). Additionally, Deeks' linear regression test indicated that there exists a publication bias in the included studies ($p = 0.002$). Thus, the results should be interpreted with caution.

Keywords: machine learning; diagnostic accuracy; mammography; meta-analysis; breast cancer



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REVIEW ARTICLE

Top 100 Most-Cited Publications on Breast Cancer and Machine Learning Research: A Bibliometric Analysis

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Abstract: Background: Rapid advancement in computing technology and digital information leads to the possible use of machine learning on breast cancer.

Objective: This study aimed to evaluate the research output of the top 100 publications and further identify a research theme of breast cancer and machine-learning studies.

Methods: Databases of Scopus and Web of Science were used to extract the top 100 publications. These publications were filtered based on the total citation of each paper. Additionally, a bibliometric analysis was applied to the top 100 publications.

Results: The top 100 publications were published between 1993 and 2019. The most productive author was Giger ML, and the top two institutions were the University of Chicago and the National University of Singapore. The most active countries were the USA, Germany, and China. Ten clusters were identified as both basic and specialised themes of breast cancer and machine learning.

Conclusion: Various countries demonstrated comparable interest in breast cancer and machine-learning research. A few Asian countries, such as China, India and Singapore, were listed in the top 10 countries based on the total citation. Additionally, the use of deep learning and breast imaging data was trending in the past 10 years in the field of breast cancer and machine-learning research.

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Keywords: Bibliometrics, breast cancer, machine learning, research trend, research output, research productivity.

1. INTRODUCTION

Breast cancer has been the most common cancer among women, which affects 25% of women worldwide [1]. The incidence of breast cancer is higher in developed than in developing countries [2]. However, the difference is most likely to be overemphasised, as there is a lack of screening in developing countries [3]. Additionally, the incidence of breast cancer was reported to steadily increase over the years in South America, Africa, and Asia [2]. A study published in 2019 esti-

Machine learning is a branch of artificial intelligence (AI). It is one of the most thriving subfield areas of AI due to advances in computer electronic and digital information. Machine learning mainly aims to make predictions and classifications based on a pattern learned from a dataset. Subsequently, this prediction and classification are used to support the decision-making process. Since 2006, deep learning or representative learning has gained popularity and has been used in many areas, including medicine and healthcare [5]. Deep learning is a

Role of Artificial Intelligence in the Early Diagnosis of Oral Cancer. A Scoping Review

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Simple Summary: Oral cancer is characterized by high morbidity and mortality, since the disease is typically in an advanced locoregional stage at the time of diagnosis. The application of artificial intelligence (AI) techniques to oral cancer screening has recently been proposed. This scoping review analyzed the information about different machine learning tools in support of non-invasive diagnostic techniques including telemedicine, medical images, fluorescence images, exfoliative cytology and predictor variables at risk of developing oral cancer. The results suggest that such tools can make a non-invasive contribution to the early diagnosis of oral cancer and we express the gaps of the proposed questions to be improved in new investigations.

Abstract: The early diagnosis of cancer can facilitate subsequent clinical patient management. Artificial intelligence (AI) has been found to be promising for improving the diagnostic process. The aim of the present study is to increase the evidence on the application of AI to the early diagnosis of oral cancer through a scoping review. A search was performed in the PubMed, Web of Science, Embase and Google Scholar databases during the period from January 2000 to December 2020, referring to the early non-invasive diagnosis of oral cancer based on AI applied to screening. Only accessible full-text articles were considered. Thirty-six studies were included on the early detection of oral cancer based on images (photographs (optical imaging and enhancement technology) and cytology) with the application of AI models. These studies were characterized by their heterogeneous nature. Each publication involved a different algorithm with potential training data bias and few comparative data for AI interpretation. Artificial intelligence may play an important role in precisely predicting the development of oral cancer, though several methodological issues need to be addressed in parallel to the advances in AI techniques, in order to allow large-scale transfer of the latter to population-based detection protocols.



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Special Editorial

A Critical Appraisal of COVID-19 in Malaysia and Beyond

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Abstract

When the first report of COVID-19 appeared in December 2019 from Wuhan, China, the world unknowingly perceived this as another flu-like illness. Many were surprised at the extreme steps that China had subsequently taken to seal Wuhan from the rest of the world. However, by February 2020, the SARS-CoV-2 virus, which causes COVID-19, had spread so quickly across the globe that the World Health Organization officially declared COVID-19 a pandemic. COVID-19 is not the first pandemic the world has seen, so what makes it so unique in Malaysia, is discussed to avoid a future coronacom.

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Premature Cardiovascular Death and its Modifiable Risk Factors: Protocol of a Systematic Review and Meta-Analysis

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ABSTRACT

Introduction: The burden of cardiovascular disease (CVDs), in the number of premature deaths, continues to increase globally and alarmingly, especially in almost all countries outside high-income countries. An accurate estimate of premature CVD death is of crucial importance for planning, implementing, and evaluating cardiovascular prevention and care interventions. However, existing literature reported the evidence on premature CVD death from selected regions, and there has been less evidence of systematic review with meta-analysis estimating the global premature death. This paper reports the protocol for a systematic review and meta-analysis to derive solid and updated estimates on global and setting-specific premature CVD death prevalence and its associated risk factors.

Methods and analysis: PUBMED, EMBASE, Web of Science, CINAHL, and Cochrane Central Register of Controlled Trials (CENTRAL) will be used as the literature database. Retrieved records will be independently screened by two authors and relevant data will be extracted from studies that report data on premature mortality (death before the age of 70 years) related to CVD. Study selection and reporting will follow the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guideline. Pooled estimates of

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Publication Types: Letter

Oral lesions in a patient with Covid-19

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Letter to Editor: Oral lesions in a patient with Covid-19

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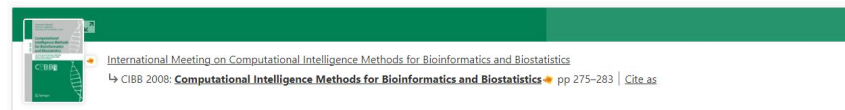
Soares CD, Carvalho RAd, Carvalho KAd, Carvalho MGFd, Almeida OPl. Oral lesions in a patient with Covid-19. *Med Oral Pathol Oral Cir Buceal.* 2020 Jul 1;25 (4):e563-4.

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Dear Editor:

The coronavirus disease 2019 (COVID-19) is a global pandemic burden caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection with variable clinical outcome. The symptoms of this disease include headache, sore throat, fever, and dyspnoea. About 10% of the patients develop severe acute respiratory syndrome and 1-2 %, particularly elderly, die (1,2). Possible oral-related symptoms include hypospasia, xerostomia and chemosensory alterations (3).

We report here the clinical and microscopic characteristics of oral reddish lesions and ulceration that occurred in a 42-year-old male patient positive for SARS-CoV-2 confirmed by polymerase chain reaction (PCR). The patient also had a history of diabetes and hypertension, and when admitted to the hospital presented fever (temperature of up to 39.3°C), cough and shortness of breath. On the skin it was observed some petechia-like and small vesiculobullous lesions of unknown aetiology. A treatment with dexamethasone and dipyrone was established for 1 week. The patient also complained of a painful ulceration in the buccal mucosa that was biopsied. Oral examination showed



International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics
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A Supervised Learning Technique and Its Applications to Computational Biology

Mario R. Guarracino, Altannar Chinchuluun & Panos M. Pardalos

Conference paper

812 Accesses

Part of the [Lecture Notes in Computer Science](#) book series (LNBI, volume 5488)

Abstract

The problem of classifying data in spaces with thousands of dimensions have recently been addressed in literature for its importance in computational biology. An example of such applications is the analysis of genomic and proteomic data. Among the most promising techniques that classify such data in lower dimensional subspace, Top Scoring Pairs has the advantage of finding a two-dimensional subspace with a simple decision rule. In the present paper we show how this technique can take advantage from the utilization of incremental generalized eigenvalue classifier to obtain higher classification accuracy with a small training set.

Keywords

[Classification](#) [Top Scoring Pair](#) [generalized eigenvalue classification](#)

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References

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ORIGINAL ARTICLE: INFECTION AND IMMUNITY



Clinical characteristics of a case series of children with coronavirus disease 2019

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Abstract

We reported the clinical characteristics of a case series of 10 patients with coronavirus disease 2019 (COVID-19) aged from 1 year to 18 years. Seven patients had contact with confirmed COVID-19 family members before onset. Fever (4 [40.0%]) and cough (3 [30.0%]) were the most common symptoms. No patient showed leucopenia and lymphopenia on admission. Pneumonia was observed in chest CT images in 5 (50.0%) patients. Five (50.0%) patients received antiviral treatment. No patient had severe complications or developed a severe illness in our study. Our study indicated that COVID-19 children present less severe symptoms and have better outcomes.

KEYWORDS

children, coronavirus disease 2019, prognosis

1 | INTRODUCTION

In December 2019, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection spread rapidly among humans worldwide.¹ The epidemiological, radiological, and clinical characteristics of COVID-19 have been reported in the general population.² Elderly patients with comorbidities may be more likely infected by SARS-CoV-2 according to previous reports.³ In addition, the characteristics of COVID-19 in special populations were also reported. Wei et al⁴ reported nine confirmed infants in Wuhan under 1 year of age with a good prognosis. However, to the best of our knowledge, few studies have reported the clinical characteristics of COVID-19 in

2 | METHODS

Children aged from 1 year to 18 years with confirmed COVID-19 were retrospectively recruited from 3 designated hospitals in three cities of Jiangsu province, China. All children were confirmed with COVID-19 between January 24, 2020 and February 22, 2020 based on the criterion of the World Health Organization (WHO) interim guidance.⁵

Demographic, epidemiological and clinical data were collected from medical records. All patients were confirmed by quantitative real-time reverse transcriptase polymerase chain reaction (RT-PCR) method on throat swab sample or anal swab sample. The protocol of

- Which one to choose?
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 - Editorial (written by an expert, usually member of editorial boards):
 - Update knowledge on the progress in the field of study
 - To get an idea to carry out research work

- Letters (written in response to an issue or article):
 - Update knowledge on the progress in the field of study
 - To get an idea to carry out research work
- Case report/case series (more common in fields such as medicine and psychology):
 - Update knowledge on the progress in the field of study
 - To get an idea to carry out research work

Where to find a paper?

- Paid databases:
 - Scopus, web of science, etc
- Free databases:
 - [Google scholar](#), [PubMed](#), [PubMed Central](#), etc
- Non-databases:
 - [ResearchGate](#)
 - [Sci-Hub](#)
 - Email the authors

Sections in a paper

- Title: topic and information about the author
- Abstract: brief overview of the paper
- Introduction:
 - Background info
 - Research objectives
- Methods:
 - How the study was conducted
 - Instruments and variables utilised
 - Analysis
- Results: data, tables, and figures
- Discussion: interpretation and implication of the result
- Conclusion: summary of the paper in a few words
- References

Original Article

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Risk Factors for COVID-19 Mortality in Malaysia

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Abstract

Background: Understanding the risks of COVID-19 mortality helps in the planning and prevention of the disease. This study aimed to determine the risk factors for COVID-19 mortality in Malaysia.

Methods: Secondary online data provided by the Ministry of Health, Malaysia and Malaysia's national COVID-19 immunisation programme were used: i) COVID-19 deaths data; ii) vaccination coverage data and iii) population estimate data. Quasi-Poisson regression was performed to determine the risk factors for COVID-19 mortality.

Results: Four risk factors were identified: i) vaccination status (partial versus unvaccinated, incidence rate ratio [IRR]: 0.59; 95% CI: 0.54, 0.64; complete versus unvaccinated, IRR: 0.50; 95% CI: 0.45, 0.56; booster versus unvaccinated, IRR: 0.13; 95% CI: 0.05, 0.26); ii) age group (19 years old–59 years old versus above 60 years old, IRR: 0.90; 95% CI: 0.84, 0.97; 13 years old–18 years old versus above 60 years old, IRR: 0.09; 95% CI: 0.04, 0.19; 6 years old–12 years old versus above 60 years old, IRR: 0.09; 95% CI: 0.03, 0.22; below 5 years old versus above 60 years old, IRR: 0.11; 95% CI: 0.04, 0.23); iii) gender (male versus female, IRR: 1.23; 95% CI: 1.14, 1.32) and iv) comorbidity (yes versus no, IRR: 2.13; 95% CI: 1.96, 2.32).

Conclusion: This study highlighted the risk factors for COVID-19 mortality and the benefit of COVID-19 vaccination, especially of booster vaccination, in reducing the risk of COVID-19 mortality in Malaysia.

Keywords: COVID-19, Malaysia, mortality, risk factor, vaccination

Critical appraisal tools

- Assess the trustworthiness, relevance and results of the published paper
- When to use?
 - Not sure of the methodological flaws of the paper
 - Commonly used in systematic review, bibliometric review and meta-analysis paper
- Each tool is specific to each study design
- Example (a lot on the internet):
 - [JB](#)
 - [CASP](#)
 - [SIGN](#)

JBICRITICAL APPRAISAL CHECKLIST FOR COHORT STUDIES

Reviewer _____ Date _____

Author _____ Year _____ Record Number _____

	Yes	No	Unclear	Not applicable
1. Were the two groups similar and recruited from the same population?	☐	☐	☐	☐
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?	☐	☐	☐	☐
3. Was the exposure measured in a valid and reliable way?	☐	☐	☐	☐
4. Were confounding factors identified?	☐	☐	☐	☐
5. Were strategies to deal with confounding factors stated?	☐	☐	☐	☐
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	☐	☐	☐	☐
7. Were the outcomes measured in a valid and reliable way?	☐	☐	☐	☐
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?	☐	☐	☐	☐
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	☐	☐	☐	☐
10. Were strategies to address incomplete follow up utilized?	☐	☐	☐	☐
11. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

JBICritical Appraisal Checklist for Case Series

Reviewer _____ Date _____

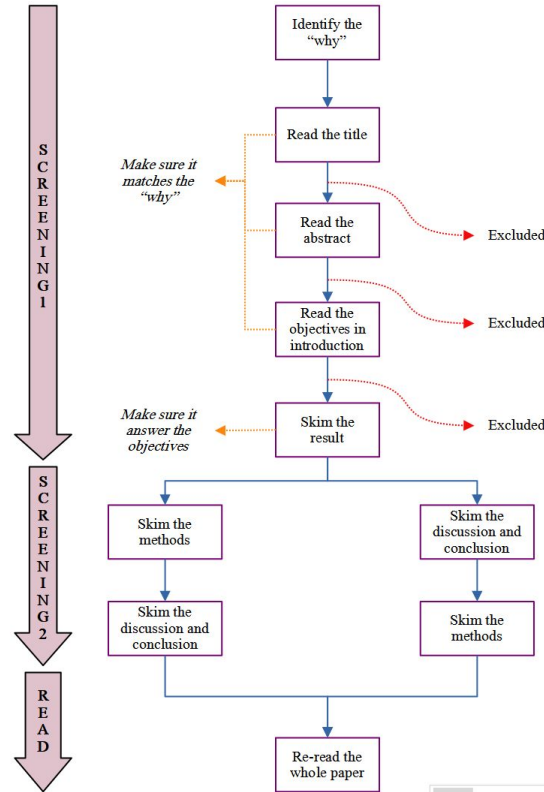
Author _____ Year _____ Record Number _____

	Yes	No	Unclear	Not applicable
• Were there clear criteria for inclusion in the case series?	☐	☐	☐	☐
• Was the condition measured in a standard, reliable way for all participants included in the case series?	☐	☐	☐	☐
• Were valid methods used for identification of the condition for all participants included in the case series?	☐	☐	☐	☐
• Did the case series have consecutive inclusion of participants?	☐	☐	☐	☐
• Did the case series have complete inclusion of participants?	☐	☐	☐	☐
• Was there clear reporting of the demographics of the participants in the study?	☐	☐	☐	☐
• Was there clear reporting of clinical information of the participants?	☐	☐	☐	☐
• Were the outcomes or follow up results of cases clearly reported?	☐	☐	☐	☐
• Was there clear reporting of the presenting site(s)/clinic(s) demographic information?	☐	☐	☐	☐
• Was statistical analysis appropriate?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

Steps (my way) in reading a paper



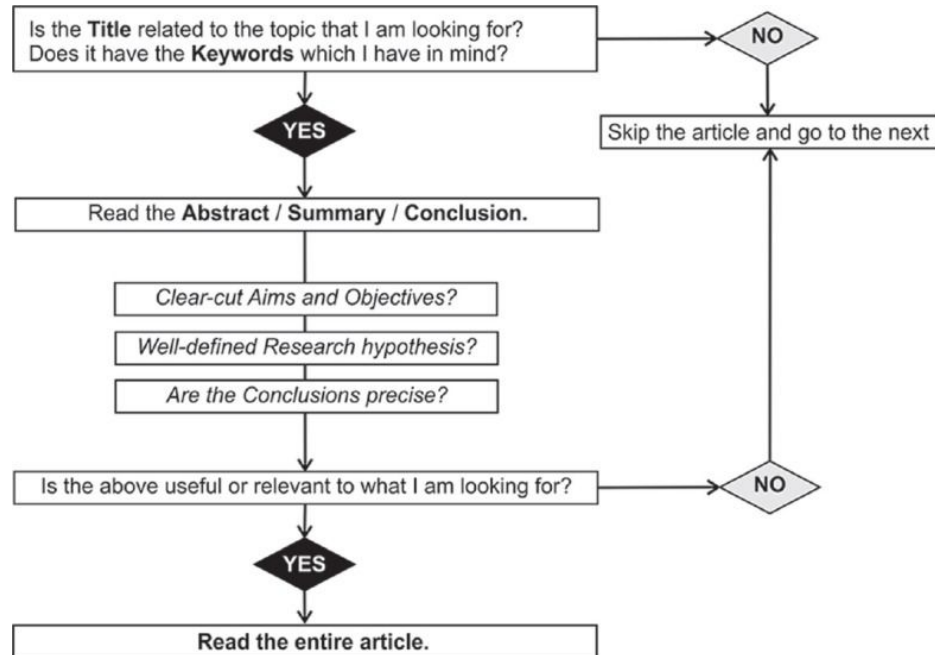
Other approaches

1. Three-pass method:

- Pass 1:
 - Read the title, abstract, and introduction
 - Get a general idea of the paper's topic, research question, and methodology
 - Identify the key terms and concepts
- Pass 2:
 - Read the paper in more detail
 - Pay attention to the figures and tables
 - Identify relevant unread reference for further reading
 - Ask yourself questions about the paper as you read it

- Pass 3:
 - Read the paper in depth
 - Try to visualise the flow of the research in the paper
 - Write down ideas for future work
 - At the end, should be able to identify the strong and weak points of the paper, related potential issue, and missing reference citation

2. How to decide whether to read a paper or not?



Tips for reading a paper

- Have a clear purpose of reading a paper
- Be critical when reading a paper - question everything
- Relate the findings to your own experience/knowledge
- Use Mendeley or any organisational tools in sorting the papers
- Make a table when reading a paper
- Not all research papers worth reading
- Adapt others' methods of reading to your own

AI tools - use or not?

- Definitely use
- Be very critical on each AI response - double and triple confirm the response
- AI tools as supporting/supplementary tools
- Do not be fully dependent on AI tools

References

- Keshav, Srinivasan. "[How to read a paper.](#)" ACM SIGCOMM Computer Communication Review 37.3 (2007): 83-84.
- Subramanyam, R. V. "[Art of reading a journal article: Methodically and effectively.](#)" Journal of oral and maxillofacial pathology: JOMFP 17.1 (2013): 65.



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